# BEING A CLOSED IMMERSION IS FPQC LOCAL ON THE BASE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- A closed immersion is affine, and affineness descends along an fpqc covering. Morphisms, Lemma morphisms-lemma-closed-immersion-affine and Same chapter, Lemma lemma-descending-property-affine, invoked at proof lines 6195-6198. Makes f affine, so that the remaining question is surjectivity of a sheaf map.
- Flat base change for the pushforward of the structure sheaf. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology, invoked at proof lines 6202-6204. Identifies the pushforward along the base change with the pullback of f_* O_X.

*Source reference.* Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 02L5: sources/stacks-descent/pinned-original-source.tex, source lines 6143–6143. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 01SE: sources/stacks-chow/tags.txt, source lines 2335–2335. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *The property $\mathcal{P}(f) =$"$f$ is a closed immersion" is fpqc local on the base.*

*Proof.* Let $f : X \to Y$ be a morphism of schemes. Let $\{Y_i \to Y\}$ be an fpqc covering. Assume that each $f_i : Y_i \times_Y X \to Y_i$ is a closed immersion. This implies that each $f_i$ is affine, see Morphisms, Lemma 01SE. By Lemma 02L5 we conclude

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that $f$ is affine. It remains to show that $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective. For every $y \in Y$ there exists an $i$ and a point $y_i \in Y_i$ mapping to $y$. By Cohomology of Schemes, Lemma 02KH the sheaf $f_{i,*}(\mathcal{O}_{Y_i \times_Y X})$ is the pullback of $f_*\mathcal{O}_X$. By assumption it is a quotient of $\mathcal{O}_{Y_i}$. Hence we see that

$$\Big(\mathcal{O}_{Y,y} \longrightarrow (f_*\mathcal{O}_X)_y\Big) \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y_i,y_i}$$

is surjective. Since $\mathcal{O}_{Y_i,y_i}$ is faithfully flat over $\mathcal{O}_{Y,y}$ this implies the surjectivity of $\mathcal{O}_{Y,y} \longrightarrow (f_*\mathcal{O}_X)_y$ as desired. $\square$